

$$H(s) = \frac{V_1}{V_2} \quad (1)$$

$$V_X = V_2 \frac{G_4}{G_4 + G_5} \quad (2)$$

$$V_X = V_X \left(1 - \frac{G_5 G_3}{G_4 G_5} \right) \quad (3)$$

$$(V_1 - V_X) G = V_X \cdot G_5 + (V_X - V_A) G_7 \quad (4)$$

$$(2) \text{ y } (3) \rightarrow (4): V_1 G - V_X G = V_X G_5 + \left(\cancel{V_X} - \cancel{V_X} + V_X \frac{G_3 G_5}{G_4 G_5} \right) \cdot G_7$$

$$V_1 \cdot G = V_X \left(G + G_5 + \frac{G_7 G_3 G_5}{G_4 G_5} \right) = V_X \left(\frac{G \cdot G_4 \cdot G_5 + G_4 G_5^2 + G_7 G_3 G_5}{G_4 G_5} \right)$$

$$V_1 \cdot G = V_2 \frac{\cancel{G_4}}{G_4 + G_5} \cdot \frac{s^2 C^2 G_4 + G_5 \cdot G \cdot G_4 + G_7 G_3 G_5}{\cancel{G_4} G_5}$$

$$\left[V_2 = V_1 \cdot G (G_4 + G_5) \cdot \frac{G_5}{s^2 C^2 \cdot G_4 + G_5 \cdot G \cdot G_4 + G_7 G_3 G_5} \right] \quad (5)$$

$$(5) \rightarrow (1): H(s) = \frac{G (G_4 + G_5) \cdot \cancel{G}}{C^2 G_4} \cdot \frac{s}{s^2 + s \frac{G}{C} + \frac{G_7 G_3 G_5}{C^2 G_4}}$$

$$H(s) = \left(\frac{1 + G_5}{G_4} \right) \cdot \frac{s \frac{G}{C}}{s^2 + s \frac{G}{C} + \frac{G_7 G_3 G_5}{C^2 \cdot G_4}}$$